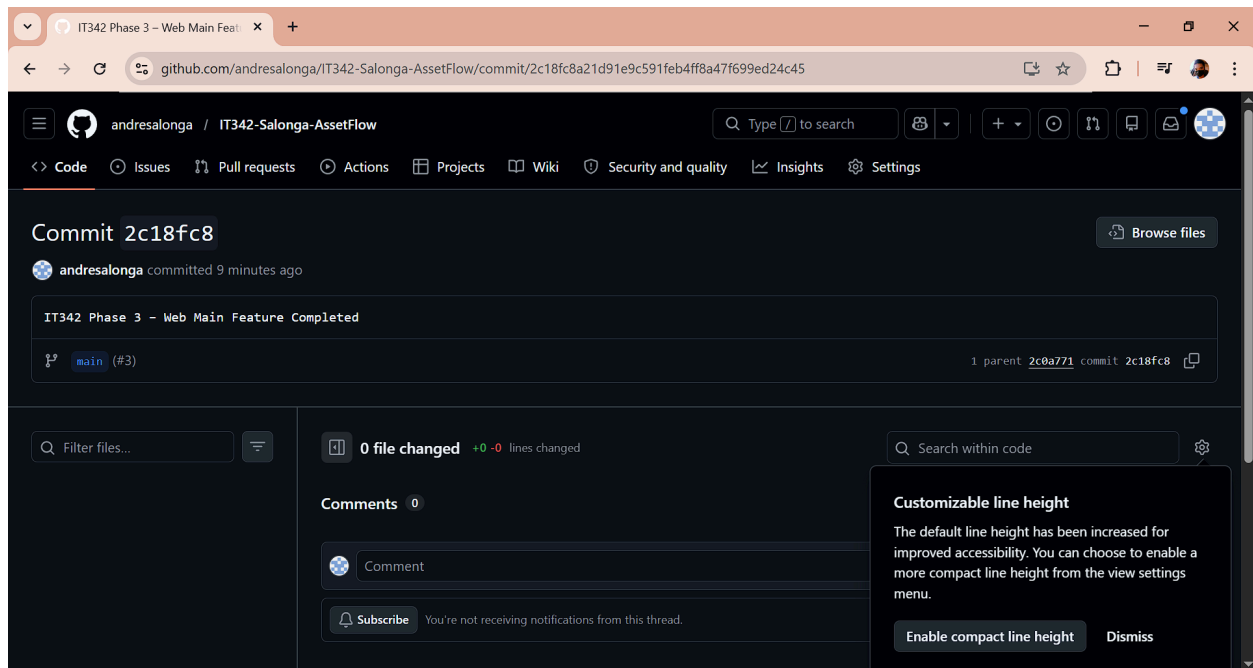


1. **GitHub Repository Link:** <https://github.com/andresalonga/IT342-Salonga-AssetFlow.git>

## 2. Final Commit for Phase 3

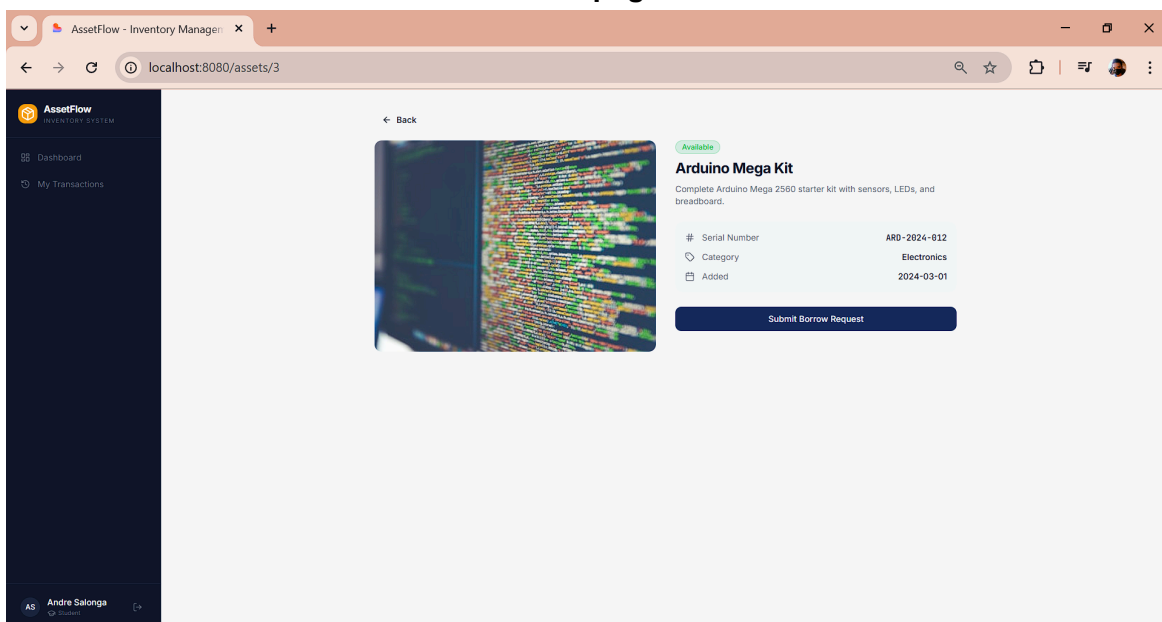
IT342 Phase 3 – Web Main Feature Completed:

<https://github.com/andresalonga/IT342-Salonga-AssetFlow/commit/2c18fc8a21d91e9c591feb4ff8a47f699ed24c45>

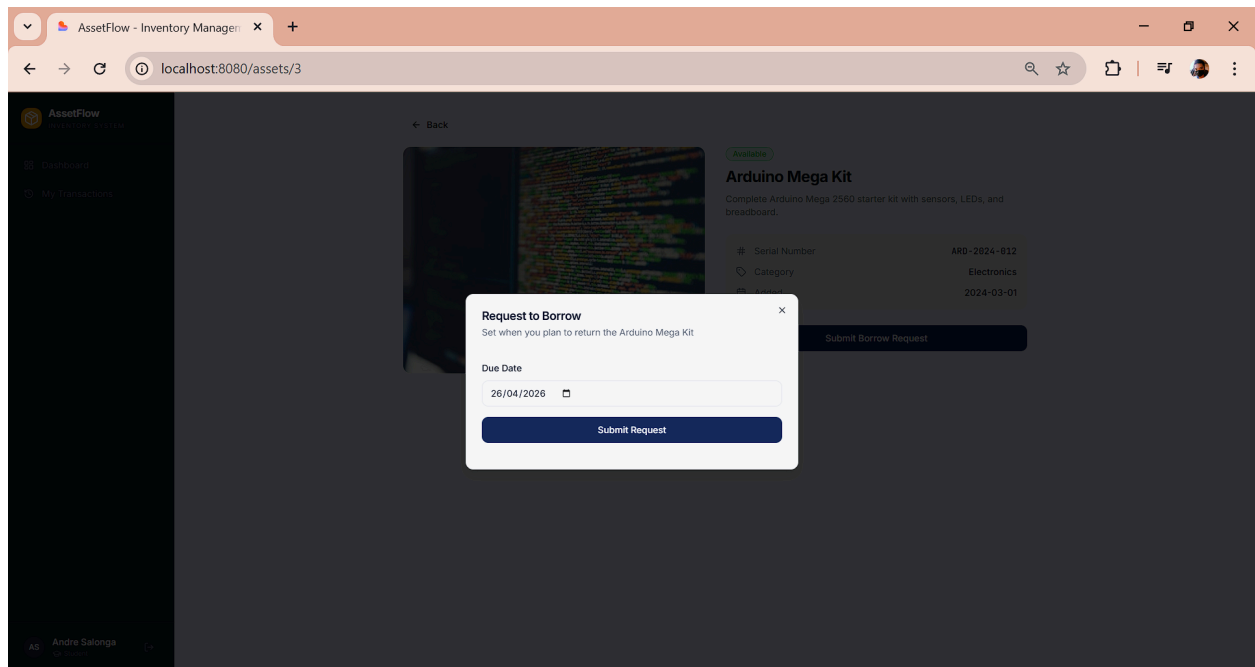


## 3. Screenshots

### Main feature page/screen

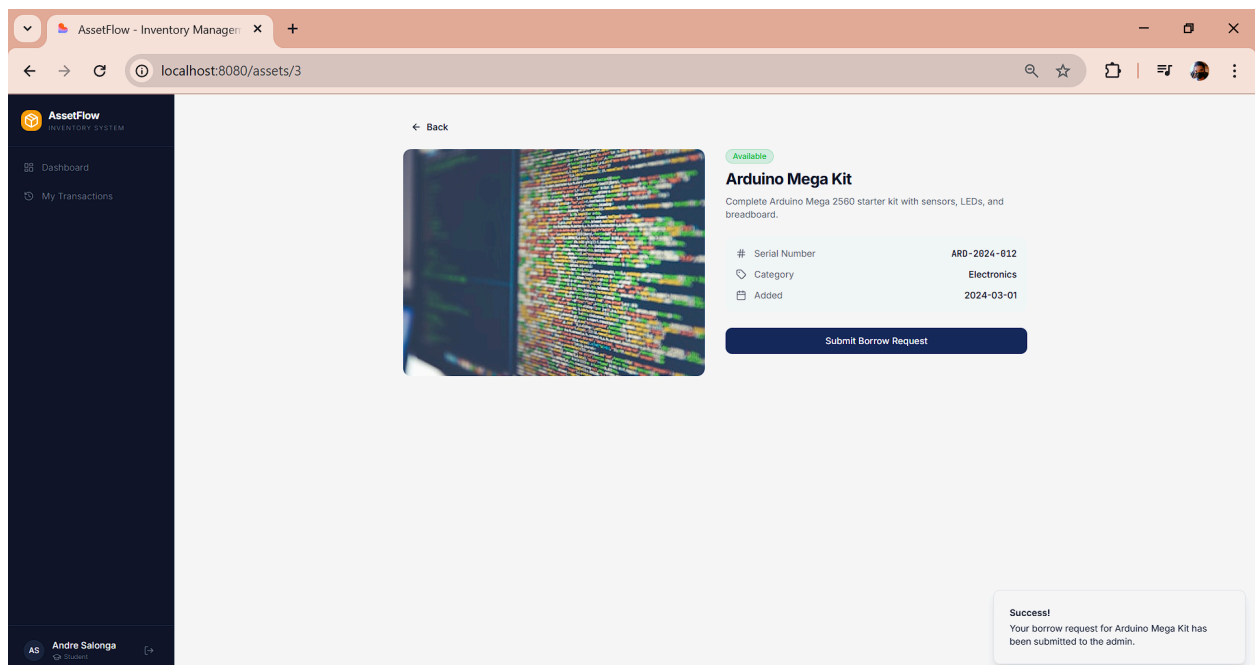


## Adding or using the main feature

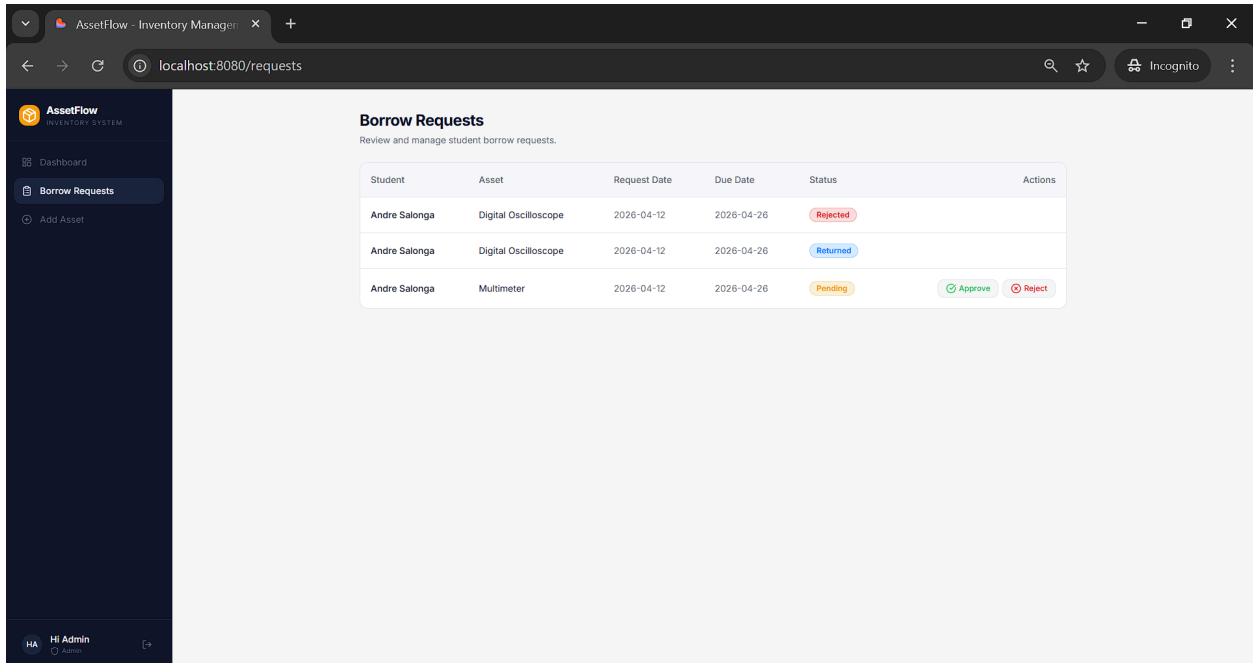


(Item Borrow Request feature. The Student has to select a return date for the item before submitting the borrow request.)

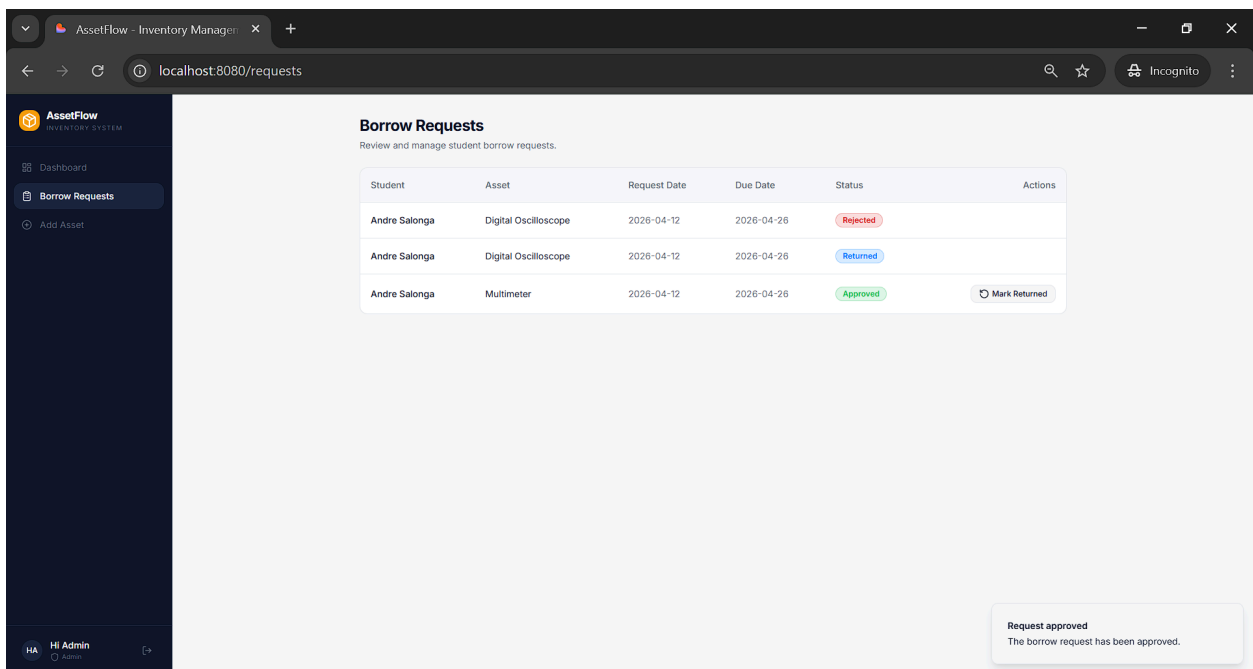
## Successful output/result



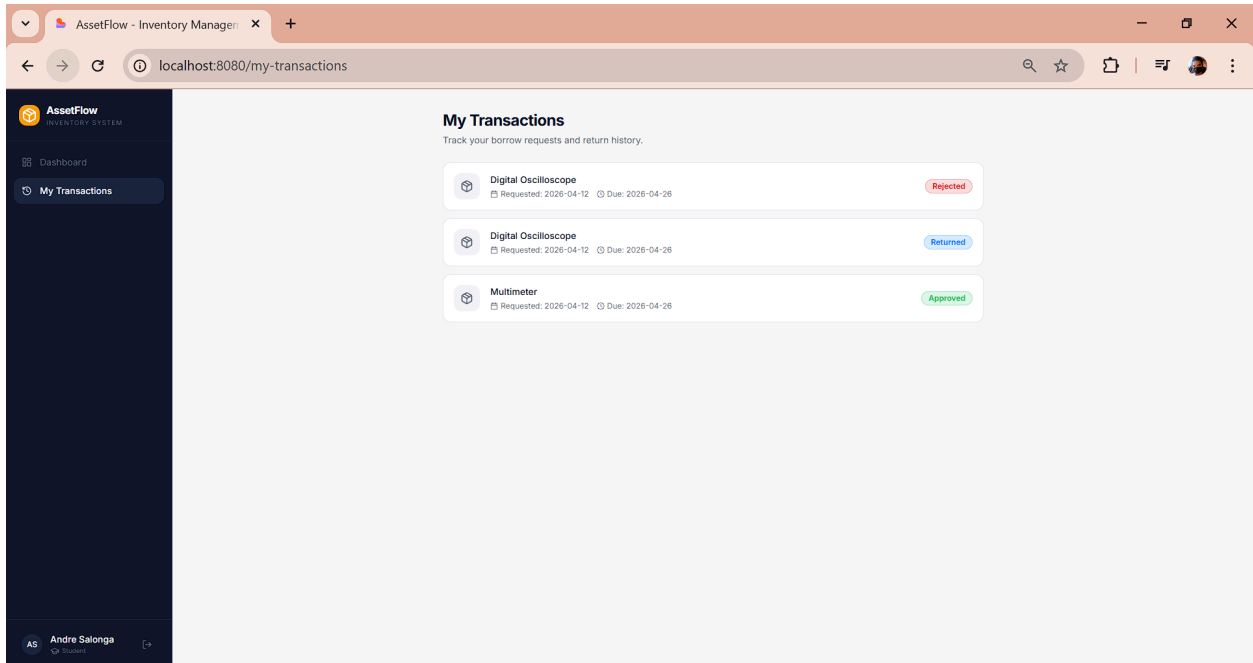
(The toast message says the borrow request was successful, and has been submitted to the Admin.)



(Admin sees the borrow request in the Borrow Requests page.)



(Admin approves the borrow request. An action 'Mark Returned' button appears, which shall be clicked by the Admin once the item has been returned by the Student.)



(In the My Transactions page of the Student, the status has been updated from 'Pending' to 'Approved.' It will change to a 'Returned' status if the item has been returned.)

## Database record related to the feature

The screenshot shows the Supabase Table Editor for the 'public.transactions' table. The table has the following columns: `id` (int8), `created_at` (timestamp), `due_date` (date), `request_date` (date), `return_date` (date), and `status` (varchar). The table contains 4 records. The third record (id=3) is highlighted, showing a status of 'APPROVED'.

|  | id | created_at                | due_date   | request_date | return_date | status   |
|--|----|---------------------------|------------|--------------|-------------|----------|
|  | 1  | 2026-04-12 08:07:36.17339 | 2026-04-26 | 2026-04-12   | NULL        | REJECTED |
|  | 2  | 2026-04-12 08:25:54.37783 | 2026-04-26 | 2026-04-12   | 2026-04-12  | RETURNED |
|  | 3  | 2026-04-12 09:27:04.57032 | 2026-04-26 | 2026-04-12   | NULL        | APPROVED |
|  | 4  | 2026-04-12 09:34:18.94051 | 2026-04-26 | 2026-04-12   | NULL        | PENDING  |

(The third record shows the approved borrow request.)

## 4. Short Summary

### Description of the Main Feature

The Borrow Request Management System is the core feature of AssetFlow. It allows students to submit borrow requests for lab equipment/assets, and administrators to review, approve, reject, or mark items as returned. The system tracks equipment availability and maintains a complete transaction history.

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### Inputs and Validations Used

- **Student Input:**
    - Asset selection (from available assets)
    - Due date selection (date picker, minimum 2 weeks from today)
    - Validation: Non-empty fields, valid date format (YYYY-MM-DD)
  - **Error Handling:**
    - 400 Bad Request: Invalid date format or missing asset
    - 403 Forbidden: Unauthenticated or unauthorized access
    - CORS security: Only authenticated requests accepted
    - Fallback to localStorage if backend unavailable
- 

### How the feature works

- 1. Student Flow:**
  - Login as student (non-admin email)
  - Browse dashboard → Click asset → Submit Borrow Request
  - Select due date → Confirm submission
  - View request status in "My Transactions" page
  - Automatically updates when admin approves/rejects
- 2. Admin Flow:**
  - Login as admin (email with "admin.com")
  - Navigate to "Borrow Requests" page
  - View all pending requests table
  - Click Approve → Asset marked as BORROWED, status updated
  - Click Reject → Request rejected, asset remains AVAILABLE
  - Click "Mark Returned" → Asset status reverted to AVAILABLE
- 3. Database Updates:**
  - Request submission creates Transaction record (PENDING status)

- Approval updates: Transaction status → APPROVED, Asset status → BORROWED
  - Return marks: Transaction status → RETURNED, Asset status → AVAILABLE, Return date recorded
- 

### API endpoints used

| Method | Endpoint                                | Purpose                   |
|--------|-----------------------------------------|---------------------------|
| POST   | /api/assets/{assetId}/borrow            | Submit borrow request     |
| GET    | /api/assets/borrow-requests             | Admin: Get all requests   |
| GET    | /api/assets/my-requests                 | Student: Get own requests |
| PATCH  | /api/assets/borrow-requests/{requestId} | Update request status     |

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### Database table/s involved

- **users** - User accounts (email, name, role, password)
- **assets** - Equipment/items (name, serial, status, category)
- **categories** - Asset categories (name, description)
- **transactions** - Borrow requests (user\_id, asset\_id, request\_date, due\_date, return\_date, status)